

Belief functions and valid statistical inference¹

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¹Thanks to friends and collaborators M. Balch, H. Crane, C. Liu...

- Belief functions have a wide range of applications.
- Just look at this conference's impressive program!
- My focus here is on classical *statistical inference*, what Dempster was thinking about when he started all this.
- Despite Dempster's genius, additive beliefs have cornered the market on uncertainty quantification in statistics.
- In this talk, I hope to convince you that, for certain concrete reasons, non-additivity is actually *necessary*.

- Statistical problem:
 - Observable data is X ;
 - Model is $X \sim P_\theta$, where $\theta \in \Theta$ is *unknown*.
- Goal is to quantify uncertainty about θ based on $X = x$.
- Various ways to approach construction of an additive belief or “posterior distribution” for θ , given x :
 - Bayes
 - fiducial/confidence
- Can also construct non-additive belief functions for θ , given x .
- To be or not to be (additive)?

- Comparison of additive and non-additive beliefs is most often treated *philosophically*, i.e., which is “right”?
- Different philosophies can co-exist, but it's perhaps easier to compare in terms of statistical properties.
- From this perspective, there is a difference!
- Take-away messages:
 - *certain statistical properties fail under additivity,*
 - *those properties can be attained under non-additivity,*
 - *and, in fact, basically anything that satisfies these good properties corresponds to a non-additive belief*

- The price of additivity:
 - satellite collision example
 - false confidence theorem
- Overcoming false confidence with valid beliefs
 - validity property
 - achieving validity via *inferential models* (IMs)
- A sort of converse to the IM validity theorem
- Concluding remarks, etc.

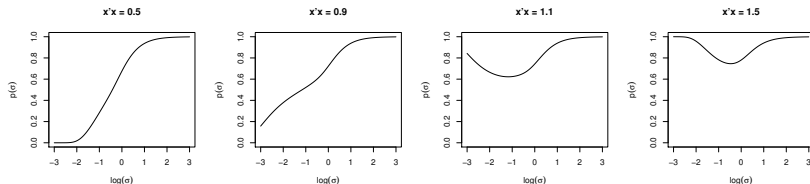
Satellite collision problem

- A satellite orbiting Earth could collide with another object.
- Potential big mess, so navigators try to avoid collision.
- Data on position, velocity, etc is used, along with physical and statistical models, to compute a *collision probability*.
- If collision probability is low, then satellite is judged safe; otherwise, some action is taken.
- An unusual phenomenon has been observed: noisier data makes collision probability small...
- Systematically misleading conclusions – bad!

Satellite collision, cont.

■ Illustration:

- $\|x\|$ denotes measured distance between satellite and object.
- True distance ≤ 1 implies collision.
- Measurement error variance is σ^2 .
- $p_x(\sigma) =$ probability of non-collision, given x .
- When σ is large, $p_x(\sigma)$ is large, for any x .
- And large σ is the reality in this application.
- Therefore, collision probability is not a reliable metric!



- What's going on here?
- Apparently, data might not be sufficiently informative with respect to questions about collision/non-collision.
- *Additivity* forces the probability to go somewhere, happens that it goes to non-collision no matter what data says.
- Even if the satellite is on a direct collision course, the probability calculations will tell navigators that it's safe.
- *False confidence*:² a hypothesis tending to be assigned large probability even though data does not support it.
- Is this general, or just something weird about this problem?

²Balch, M., and Ferson, arXiv:1706.08565

False confidence theorem.

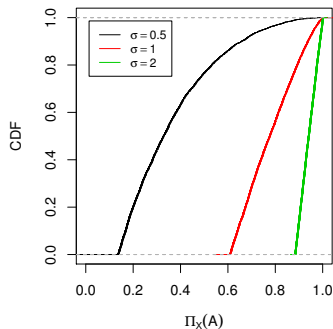
Let Π_X be any additive belief on Θ , depending on data X . Then, for any α and any t , there exists $A \subset \Theta$ such that

$$A \not\ni \theta \quad \text{and} \quad P_\theta\{\Pi_X(A) > t\} \geq \alpha.$$

- In words, there always exists a false hypothesis that tends to be assigned high posterior probability.
- If judgments about the plausibility of a hypothesis are made based on the magnitude of its probability, then the theorem says there's risk of systematic error.
- Reveals a practical price attached to additivity.
- Theorem doesn't say which hypotheses are problematic, or to what extent, only that they exist.

False confidence, cont.

- Toy version of the satellite example.³
 - Let A denote the non-collision hypothesis.
 - Then $\Pi_X(A)$ as a random variable, with a CDF
 - Plot CDF when data are generated under a collision course.
- *False confidence*: $\Pi_X(A)$ is almost always large!



³More examples in Carmichael and Williams, arXiv:1807.06217

- Theorem says *every additive belief function* is afflicted with false confidence.
- To avoid false confidence, we must go non-additive.
 - Let bel_x be a data-dependent belief function.
 - $\text{bel}_x(\emptyset) = 0$, $\text{bel}_x(\Theta) = 1$, and completely monotone.
 - In particular, $\text{bel}_x(A) + \text{bel}_x(A^c) \leq 1$.
- But not every non-additive belief would be satisfactory.
- That is, to avoid false confidence, the belief must satisfy some additional conditions...

- First a question: *Why should we avoid false confidence?*
- What about all the success stories of, say, Bayesian inference?
- Key point: a Bayesian posterior is good for some tasks and not for others.⁴
- But statisticians are in the business of developing *methods*.
- If my method is touted as giving a full posterior distribution, then people will want to use it as such.
- If I can't control what hypotheses a user might plug in, then I better be sure that there's no false confidence.

⁴See Grünwald's recent work on *safe probability*, arXiv:1604.01785.

Definition.

A data-dependent belief function bel_x is said to be *valid* if

$$\sup_{\theta \notin A} P_{\theta} \{ \text{bel}_x(A) > 1 - \alpha \} \leq \alpha, \quad \forall A \subseteq \Theta, \quad \forall \alpha \in (0, 1).$$

- Validity property⁵ implies that assigning high belief to a false hypothesis is a rare event — *no false confidence*.
- α on inside and outside calibrates beliefs.
- Definitions are fine, but how to achieve validity?

⁵First version of this is in M. and Liu, arXiv:1206.4091; see, also, the “calibrated beliefs” in Denoeux & Li (IJAR 2018).

- The *inferential model* (IM) approach relies on random sets.
- Express the statistical model as

$$X = a(\theta, U), \quad U \sim P_U, \quad P_U \text{ known.}$$

- Difference between IMs and fiducial:
 - Fiducial converts distribution of U to a distribution of θ .
 - IMs care about the *particular value* of U corresponding to observed data $X = x$.
 - This special value of U is unobservable, but we can “guess” its value with a random set $\mathcal{S} \sim P_{\mathcal{S}}$ in the U -space.
- There is theory to guide the choice of $P_{\mathcal{S}}$.

- Given $\mathcal{S} \sim P_{\mathcal{S}}$, push it forward to the θ space:

$$\Theta_x(\mathcal{S}) = \bigcup_{u \in \mathcal{S}} \{\theta : x = a(\theta, u)\}.$$

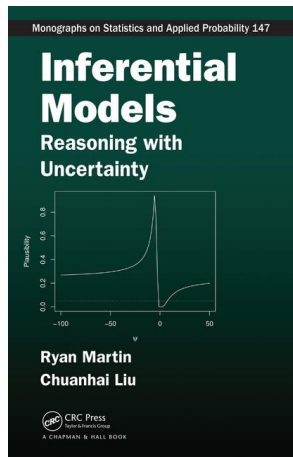
- Then compute the random set probability⁶

$$\text{bel}_x(A) = P_{\mathcal{S}}\{\Theta_x(\mathcal{S}) \subseteq A\}.$$

- This gives a genuinely non-additive belief function, along with the corresponding plausibility function

$$\text{pl}_x(A) = P_{\mathcal{S}}\{\Theta_x(\mathcal{S}) \cap A \neq \emptyset\}.$$

⁶Here I'm ignoring the possibility that $\Theta_x(\mathcal{S}) = \emptyset$.



IM validity theorem.

With a “suitable choice of random set $\mathcal{S}$,” the IM satisfies the validity property and, therefore, avoids false confidence.

- Re: “suitable choice of $\mathcal{S}$.”
 - Let $f(u) = P_{\mathcal{S}}(\mathcal{S} \ni u)$, contour function.
 - Need $f(U) \geq_{\text{st}} \text{Unif}(0, 1)$ when $U \sim P_U$.
- This is actually easy to arrange...
- Consequences of validity:
 - “Reject $H_0 : \theta \in A$ if $\text{pl}_x(A) \leq \alpha$ ” is a size α test.
 - Nominal coverage probability of $100(1 - \alpha)\%$ plausibility region

$$\mathcal{P}_{\alpha}(x) = \{\vartheta : \text{pl}_x(\{\vartheta\}) \geq \alpha\}.$$

- Let $X \sim \text{Pois}(\theta)$, distribution function F_θ .
- Construct an IM for θ :
 - A. $F_\theta(X - 1) \leq U < F_\theta(X)$, $U \sim \text{Unif}(0, 1)$.
 - P. “Default” predictive random set

$$\mathcal{S} = \{u : |u - 0.5| \leq |\tilde{U} - 0.5|, \tilde{U} \sim \text{Unif}(0, 1)\}.$$

- C. Combine to get⁷

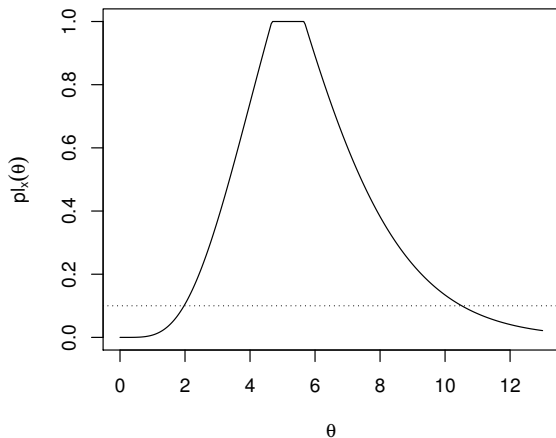
$$\begin{aligned}\Theta_x(\mathcal{S}) &= \bigcup_{u \in \mathcal{S}} \{\theta : F_\theta(x - 1) \leq u < F_\theta(x)\} \\ &= [G_x^{-1}(0.5 - |\tilde{U} - 0.5|), G_{x+1}^{-1}(0.5 + |\tilde{U} - 0.5|)].\end{aligned}$$

- Can evaluate belief and plausibility based on $\tilde{U} \sim \text{Unif}(0, 1)$.

⁷Use fact that $F_\theta(x) = 1 - G_{x+1}(\theta)$, gamma distribution.

Poisson illustration, cont.

- Observation $X = 5$.
- Plot of $pl_5(\{\theta\})$ over a range of θ .



Curved normal example

- $Y = (Y_1, \dots, Y_n)$ iid $N(\theta, \theta^2)$.⁸
- A classical non-regular problem, $\dim(\theta) < \dim(\text{suff stat})$.
- Obvious association with sufficient statistics:

$$\bar{X} = \theta + |\theta|U_1 \quad S = |\theta|U_2.$$

- Two auxiliary variables for one parameter, inefficient.⁹
- Rewrite association as

$$\bar{X} = \theta + S \frac{U_1}{U_2} \quad \text{and} \quad \frac{\bar{X}}{S} = \frac{\text{sign}(\theta) + U_1}{U_2}.$$

- Function of (U_1, U_2) is *observed*.

⁸Without (much) loss of generality, assume $\text{sign}(\theta)$ is known.

⁹Dimension-reduction strategies in M. and Liu, arXiv:1211.1530.

- Make a change-of-variable and condition on what's observed.
- New association, A-step,

$$\bar{X} = \theta + S V,$$

where V has the conditional distribution of U_1/U_2 , given...

- For P-step, let f be (conditional) density of V , and take S to be the random level sets

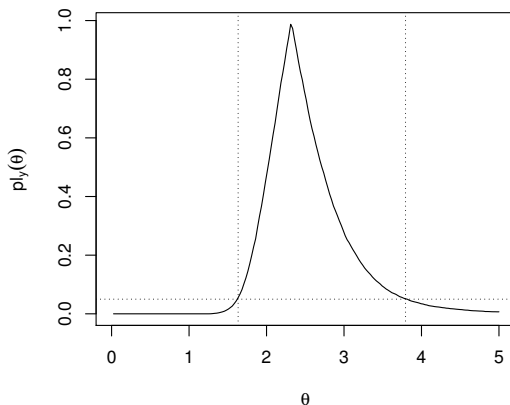
$$S = \{v : f(v) \geq f(\tilde{V}), \tilde{V} \sim f\}.$$

- C-step is just like before and, e.g.,

$$\text{pl}_X(\theta) \approx \frac{1}{M} \sum_{m=1}^M 1\{f(\frac{\bar{X}-\theta}{S}) \geq f(\tilde{V}_m)\}, \quad \tilde{V}_m \sim f.$$

Curved normal example, cont.

- Experiment: $n = 10$ and $\theta = 2$.
- Data: $\bar{X} = 1.97$ and $S = 2.48$.
- Plausibility function $pl_x(\theta)$ and 95% plausibility interval.



- IM is valid by construction, but what about efficient?
- Compare to a (generalized) fiducial solution which is (at least) first-order accurate asymptotically.
- Repeat experiment above; table shows estimated coverage probability and mean lengths of 95% intervals.

	Coverage probability	Mean length
Fiducial	0.932	1.45
IM	0.956	1.82

- Fiducial seems to be slightly shorter, but it under-covers.

- Let $X = (X_1, \dots, X_n)$ be iid $N(\mu, \sigma^2)$, $\theta = (\mu, \sigma)$.
- Goal is inference on $\psi = \sigma/\mu$, coefficient of variation.
- μ in the denominator makes this a difficult problem.
- IM approach:
 - builds a reduced-association:¹⁰

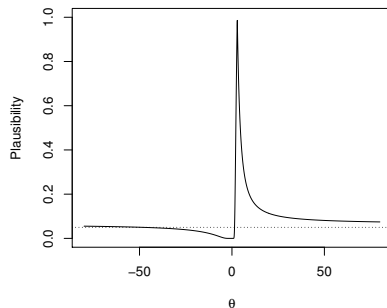
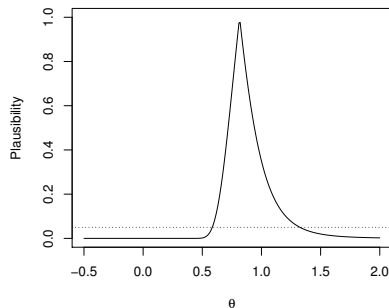
$$\frac{n^{1/2}\bar{X}}{S} = F_{n,1/\psi}^{-1}(U), \quad U \sim \text{Unif}(0,1).$$

- this yields a (marginal) IM for ψ
 - automatically valid and gives very reasonable results
- Full details are slightly too lengthy to present here...

¹⁰Dimension-reduction for marginalization in M. and Liu, [arXiv:1306.3092](#).

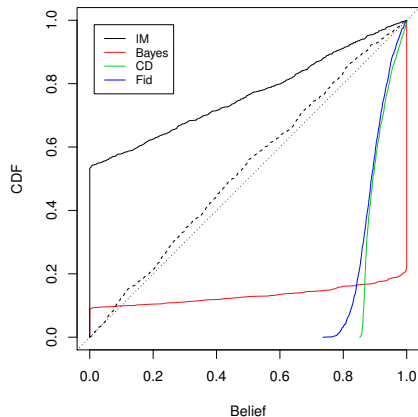
Normal CV example, cont.

- Simulate data of size $n = 30$ from $N(\mu, \sigma^2)$, with $\sigma = 1$.
- Plot of plausibility function for ψ ; based on “default” $\mathcal{S}$.
- Left: $\mu = 1$; right: $\mu = 0$.



Normal CV example, cont.

- Simulation: $n = 10$ from $N(0.1, 1)$, so that $\psi = 10$.
- Consider $A = (-\infty, 9]$, which *does not* contain $\psi = 10$.
- Compute CDFs of Bayes, fiducial, CD, and IM beliefs.¹¹



¹¹My fiducial and CD approaches here are “naive” ...

- Valid beliefs yield nominal confidence regions.
- Interesting reversal:
 - Given a nested family of confidence regions, create a function by stacking up its contours
 - Function has the shape a plausibility function and, in fact, it corresponds to a valid belief function.
- So, valid belief functions are fundamental to statistics.
 - We always tell students that confidence isn't probability, but we never tell them what confidence IS.
 - Confidence is belief/plausibility: *a confidence set contains all parameter values that are sufficiently and justifiably plausible based on the observed data.*
 - I teach my students this explanation.¹²

¹²M., arXiv:1606.02352

- So far, we have seen that
 - the IM approach is a way to construct valid beliefs,
 - and confidence regions correspond to valid beliefs.
- Not clear, however, if every confidence region corresponds to a valid belief that can be constructed via an IM.
- That is, there could be a gap between “good” beliefs and those that can be reached via IMs.
- But it turns out that there is no gap:
 - (basically) every nominal confidence region can be derived from a (“glued”) IM,
 - and there’s an algorithm for constructing the IM.
- This is a sort of converse to the IM validity theorem.

Quick interlude about random sets

- A wide class of random sets $\mathcal{S}$ that all lead to a valid IM are those we call *admissible*, i.e.,
 - the support $\mathbb{S}$ of $\mathcal{S}$ is nested, and
 - $P_{\mathcal{S}}$ satisfies $P_{\mathcal{S}}(\mathcal{S} \subseteq K) = \sup_{S \in \mathbb{S}: S \subseteq K} P_U(S)$.
- By restricting to this class, we see that the random set is completely determined by its support $\mathbb{S}$ (and the measure P_U).
- So if the goal is to construct a valid IM that satisfies such-and-such condition, then we only need to construct a suitable support.

- There is no reason that one has to use a single $\mathcal{S}$.
- Consider $\{(\mathbb{S}_\vartheta, P_{\mathcal{S}|\vartheta}) : \vartheta \in \Theta\}$, admissible for each ϑ .
- Do IM construction to get a ϑ -dependent plausibility

$$\text{pl}_x(\theta \mid \vartheta) = P_{\mathcal{S}|\vartheta}\{\Theta_x(\mathcal{S}) \ni \theta\}, \quad \theta \in \Theta.$$

- Define a “glued IM” with plausibility

$$\text{pl}_x(A) = \sup_{\vartheta \in A} \text{pl}_x(\vartheta \mid \vartheta).$$

- Validity of this glued IM follows immediately from that of each ϑ -specific IM.

- This glued IM idea may seem a bit strange.
- Has appeared in a different form in some of my papers, under the name of a “local conditional IM.”
- This additional flexibility is especially useful for reducing the dimension of U in certain non-regular cases:
 - a conditioning operation helps improve efficiency, but may not be able to find a conditioning event free of θ ;
 - roughly, our idea is to absorb the parameter dependence in the conditioning event into the family of random sets.
- No time for an example here... :(

- From a consonant plausibility on Θ , we can get a *marginal plausibility* for $\psi = \psi(\theta)$ via optimization:

$$\text{mpl}_x(\psi) = \sup_{\theta: \psi(\theta) = \psi} \text{pl}_x(\theta).$$

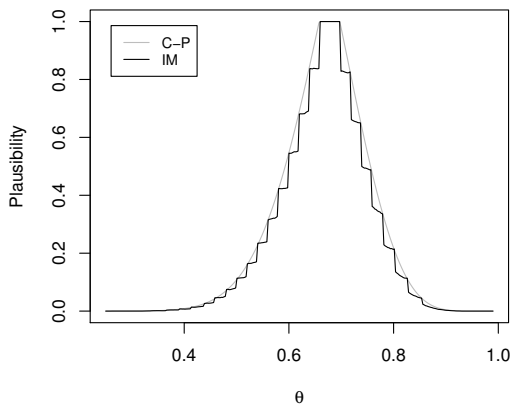
- Marginal plausibility region: $\mathcal{P}_\alpha(x) = \{\psi : \text{mpl}_x(\psi) > \alpha\}$.

Theorem (M. arXiv:1707.00486).

Given nominal confidence regions C_α for $\psi = \psi(\theta)$ that are nested and satisfy a *compatibility condition*, there exists a valid (glued) IM on Θ such that the marginal plausibility region above satisfies

$$\mathcal{P}_\alpha(x) \subseteq C_\alpha(x) \quad \text{for all } (\alpha, x).$$

- Binomial example: $n = 25$ and $x = 17$.
 - Start with Clopper–Pearson intervals, $C_\alpha(x)$.
 - Deduced plausibility region from glued IM is smaller!



- Idea of the proof:

- use C_α 's to construct the random set's support, i.e.,

$$S_\alpha(\vartheta) = \{u : C_\alpha(a(\vartheta, u)) \ni \psi(\vartheta)\}, \quad \alpha \in (0, 1), \quad \vartheta \in \Theta$$

- yields a (glued) IM on Θ , with consonant plausibility;
- marginalize to ψ via optimization.

- Compatibility condition:

- Special case where C_α depends only on minimal sufficient statistic X of the same dimension as θ .
- Need “ $x = a(\theta, u)$ ” such that $S_\alpha(\vartheta) \neq \emptyset$ for all (ϑ, α) .

- Therefore: *every good confidence region corresponds to a valid (glued) IM.*

- Non-additive beliefs are fundamental to statistics.
- That is, additive beliefs
 - put user at risk of false confidence
 - lack a direct connection to relevant statistical properties.
- I focused on connection to confidence but a similar things can be said about p-values and IM plausibilities.¹³
- Main drawback of IM approach is that it's not always easy to do; we still just don't understand it that well.
- Lots of interesting questions that remain to be answered:
 - marginalization mysteries
 - “optimal” random sets
 - incorporating prior info without losing validity
 - ...

¹³M. and Liu, arXiv:1211.1547.

Thank you!

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